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Personal Statement

Applied Robotics and AI Engineer with a PhD in Engineering Product Development and experience spanning robotics research and applied AI. Skilled in perception, localization, sensor fusion, computer vision, and machine learning, with proficiency in C++ and Python for robotics and AI applications. Experienced in developing and integrating end-to-end intelligent systems, from algorithm design and AI model development to system integration, testing, and real-world deployment. Comfortable collaborating across multidisciplinary teams.

Work Experience

Singapore Institute of Technology

Applied AI Engineer (Team Lead), Jun 2026 – Present

- Completed intensive training in machine learning, deep learning, generative AI, and MLOps.
- Developed end-to-end AI applications using LLMs, RAG, agentic workflows, Docker, Airflow, MLflow, and CI/CD pipelines.
- Led a team in building and deploying industry-focused AI solutions, contributing across the full stack including AI/ML, UI, data storage, AWS Bedrock, S3, and production deployment.
- Led development of a human-in-the-loop PPE violation detection using LLM-based visual reasoning to verify model detections and reduce false positives.

Singapore University of Technology and Design

Graduate Researcher, Sep 2020 – Aug 2025

- Developed multi-sensor fusion algorithms using UWB, LiDAR, and odometry for multi-robot localization.
- Designed perception and computer vision solutions for object detection and tracking using deep learning and geometric methods.
- Developed UWB signal processing and NLOS mitigation using machine learning.
- Conducted end-to-end research including data collection, algorithm development, real-world experimentation, and publication.
- Collaborated with DSO on localization for autonomous search-and-rescue applications.

Education

- **Singapore University of Technology and Design**
Doctor of Philosophy: Engineering Product Development, Sep 2020 - Aug 2025
- **Singapore Institute of Technology** in partnership with University of Glasgow
Bachelor of Engineering with Honours of the First Class: Mechatronics, Aug 2018 – Aug 2020
- **Temasek Polytechnic**
Diploma in Aerospace Electronics, Apr 2013 – May 2016

Skills

- **Programming:** C++, Python, ROS, Linux development
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, Computer Vision, LLMs, RAG, AI Agents, SFT/LoRA, MLOps
- **Robotics & Perception:** SLAM, Localization, Sensor Fusion, UWB, LiDAR, IMU, cameras, autonomous/mobile robots
- **Cloud & Deployment:** AWS, EC2, S3, Docker, MLflow, Airflow, CI/CD, GitHub, Jira
- **Data & Experimentation:** SQL, Pandas, NumPy, SciPy, Matplotlib, prototyping, testing, validation

Awards & Certifications

- NVIDIA DLI: Building AI Agents with Multimodal Models, Adding Knowledge to LLMs, Generative AI with Diffusion Models, Aug 2026
- Certificate of Appreciation as Invited Speaker (RAIIC), Aug 2023
- SIT-Keppel Gold Medal, Oct 2020
- Academic Achievement Award (Mendaki), Sep 2020
- Anugerah Cemerlang Mendaki (University), Sep 2020
- University of Glasgow Engineering Excellence List, Aug 2019 and Sep 2020
- SUTD President's Graduate Fellowship, Sep 2020

Publications

- [1] Localization through mitigating and compensating UWB NLOS ranging error with neural network, Elsevier DSP, 2025
- [2] MR-FLOUR: Multi-Robot Relative Localization based on the Fusion of LiDAR, Odometry, and UWB Ranging, Elsevier Mechatronics, 2025
- [3] Distributed multi-robot potential-field-based exploration with submap-based mapping and noise-augmented strategy, Elsevier RAS 2024
- [4] A Robotic System of Systems for Human-Robot Collaboration in Search and Rescue Operations, AIM 2023
- [5] Moving object localization based on the fusion of ultra-WideBand and LiDAR with a mobile robot, ROBIO 2023
- [6] GMC-pos: Graph-based multi-robot coverage positioning method, ROBIO 2023
- [7] Distributed ranging slam for multiple robots with ultra-wideband and odometry measurements, IROS 2022
- [8] NLOS ranging mitigation with neural network model for UWB localization, CASE 2022
- [9] Efficient WiFi LiDAR SLAM for autonomous robots in large environments, CASE 2022